Week 10 teacher guide

Scale, proportions, circles, and review

Keep this guide beside PowerPoint slides 569-630. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 064. Saturday | pages 251-256 | slides 569-587
Scale and proportional relationships

Day 065. Sunday | pages 257-267 | slides 588-604
Circles and first previews

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Scale, proportions, circles, and review

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Weekly check

This week’s idea

A fixed multiplier connects scaled lengths and proportional quantities. Circle formulas use radius or diameter in different ways. The final weekdays revisit the entire program.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Scale and proportional relationships

Day 064, pages 251-256 | PowerPoint slides 569-587

Goal: Use scale factors and identify relationships with a constant rate per one.

Set out: grid paper, ruler, paper rectangles.

Check first: Ratios, unit rates, rectangle area, and coordinate graphs.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Uses the same multiplier on every side. Why it matters: Preserves the shape and angles. Tutor move: Ask for a ratio for each matching side.

Squares the scale factor for area. Why it matters: Accounts for two dimensions. Tutor move: Count rows and columns of original copies.

Tests ratios or the zero point. Why it matters: Both can disprove proportionality. Tutor move: Accept either clear counterexample.

Likely misconceptions

Adds the same amount to every length. Why: Confuses additive change with scaling. Tutor move: Compare each copy-to-original ratio.

Uses the length factor for area. Why: Counts one changing direction. Tutor move: Build an array of scaled unit squares.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 569. Week 10, Saturday

Topic: Scale and proportional relationships
Student: Watch and discuss

Say: Use scale factors and identify relationships with a constant rate per one.

Ask: What do you already know about this idea?

Expect: Ratios, unit rates, rectangle area, and coordinate graphs.

If stuck: Ask for a ratio for each matching side.

Slide 570. Today

Topic: Scale and proportional relationships
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A scaled copy multiplies every length by the same factor. Proportional relationships have the form $y = kx$.

If stuck: Read the learning goal together.

Slide 571. Recall | 4 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 251 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 16 square cm; 2: LCM 12; 3: 2; 4: 60 square cm

If stuck: Give time to draw before giving a hint.

Slide 572. Recall: answers | 3 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 251 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 16 square cm; LCM 12; 2; 60 square cm

If stuck: Rebuild one missed item with its model.

Slide 573. Whole-number scale factor | 8 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 252 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: Does the same multiplier connect every pair of matching lengths?

Expect: P1: 8 cm by 12 cm; area 24 to 96 square cm, factor 4

If stuck: P1: Find the multiplier for one side and test the other sides. For area, account for both changing dimensions.

Slide 574. Fractional scale factor | 3 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 252 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: P2: Does the same multiplier connect every pair of matching lengths? P3: What real distance does one drawing unit represent?

Expect: P2: 3, 4, 5 cm; corresponding angles stay equal; P3: 90 km

If stuck: P2: Find the multiplier for one side and test the other sides. For area, account for both changing dimensions. P3: Write the units beside the scale. Multiply or divide so the desired unit remains.

Slide 575. The model | 3 min

Topic: Scale and proportional relationships

Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: The straight line y = 2x passes through the origin.

If stuck: Ask for a ratio for each matching side.

Slide 576. Area multiplier is square of length multiplier | 3 min

Topic: Scale and proportional relationships

Student: Day 064 · Workbook p. 253 · P4, P5

Say: Read P4 and P5. Give time to try before comparing methods.

Ask: P4: Does the same multiplier connect every pair of matching lengths? P5: Is output per one input constant? What happens at an input of zero?

Expect: P4: 45 square cm; P5: k = 2.5; y = 2.5x

If stuck: P4: Find the multiplier for one side and test the other sides. For area, account for both changing dimensions. P5: Compare y divided by x for two nonzero pairs, or check whether the graph passes through the origin.

Slide 577. Graph through origin versus nonzero intercept | 3 min

Topic: Scale and proportional relationships

Student: Day 064 · Workbook p. 253 · P6

Say: Read P6. Give time to try before comparing methods.

Ask: P6: Is output per one input constant? What happens at an input of zero?

Expect: P6: A: y = 2x. B is not proportional because it starts at 3 when x = 0.

If stuck: P6: Compare y divided by x for two nonzero pairs, or check whether the graph passes through the origin.

Slide 578. Non-proportional story with fixed fee | 3 min

Topic: Scale and proportional relationships

Student: Day 064 · Workbook p. 254 · P7, P8

Say: Read P7 and P8. Give time to try before comparing methods.

Ask: P7: Is output per one input constant? What happens at an input of zero?

Expect: P7: c = 3m + 4; not proportional; P8: L = 1.5t, or L = $(\frac{3}{2})$t

If stuck: P7: Compare y divided by x for two nonzero pairs, or check whether the graph passes through the origin.

Slide 579. Watch the thinking | 3 min

Topic: Scale and proportional relationships
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 6 by 10 cm; area 60 square cm

If stuck: Return to the matching unit in the model.

Slide 580. Your turn | 2 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 254 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 4; 3.5; 3.5

If stuck: Compare with the previous example without copying its numbers.

Slide 581. Two ways to see it | 8 min

Topic: Scale and proportional relationships
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Partner compare: Where does the squared area factor appear in your drawing? Agree or disagree: Equal changes in a table are enough to prove proportionality.

Expect: Accept drawings and equations that preserve the same quantities and units. Squares the scale factor for area.

If stuck: Count rows and columns of original copies.

Slide 582. A claim to check | 7 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 255 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: The area is multiplied by 9 because both dimensions are tripled.

If stuck: Tile a 3 by 3 array of original rectangles.

Slide 583. Words and meaning | 4 min

Topic: Scale and proportional relationships
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: A factor of 2 doubles every length. A 2 by 3 rectangle becomes 4 by 6. For y = 4x, k = 4. At $3 each, cost = 3 × item count.

If stuck: Read one definition and draw its example.

Slide 584. In your own words | 3 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 251 · S

Say: Use your own words and one example.

Ask: Partner compare: Where does the squared area factor appear in your drawing?

Expect: A scaled copy multiplies every length by the same factor. Proportional relationships have the form y = kx.

If stuck: Point to the model and name what each part represents.

Slide 585. Exit ticket | 8 min

Topic: Scale and proportional relationships
Student: Day 064 · Workbook p. 256 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: No. Ratios 5, $\frac{11}{3}$, $\frac{17}{5}$ differ. The rule y = 3x + 2 gives y = 2 at zero.

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 586. The week ahead | 5 min

Topic: Scale and proportional relationships
Student: Days 066-070 · Workbook pp. 268-277

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 587. Tutor reference

Topic: Scale and proportional relationships

Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: No. Ratios 5, $\frac{11}{3}$, $\frac{17}{5}$ differ. The rule y = 3x + 2 gives y = 2 at zero.

Got it: Says no and shows unequal y-to-x ratios or the nonzero intercept 2.

Almost there: Says no with a correct idea but incomplete ratio calculation or explanation.

Needs re-teaching: Says yes because both columns increase by constant amounts.

Sunday. Circles and first previews

Day 065, pages 257-267 | PowerPoint slides 588-604

Goal: Find circle measures and connect percent change and signed steps to earlier models.

Set out: round lid, string, ruler, number line.

Check first: Area, squares, percent of a whole, and negative-number order.

Pacing: Circles 35 (launch 5, explore 12, discuss 10, synthesis 3, exit 5); previews 15; final paper review 15. Total 65 min.

Explore: what to watch for

Marks the center and halves the diameter. Why it matters: Keeps the area input correct. Tutor move: Ask which segment runs from center to circle.

Names linear versus square units. Why it matters: Separates circumference from area. Tutor move: Trace the edge, then shade the interior.

Uses a percent table or a tape to track the new whole. Why it matters: Both models can show why changes do not cancel. Tutor move: Accept either and label the starting amount for each step.

Likely misconceptions

Squares the diameter. Why: Uses the given number without identifying it. Tutor move: Mark two equal radii along the diameter.

Writes $\pi = 3.14$ exactly. Why: Confuses approximation with an exact constant. Tutor move: Keep the π form and label the decimal answer about.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 588. Week 10, Sunday

Topic: Circles and first previews
Student: Watch and discuss

Say: Find circle measures and connect percent change and signed steps to earlier models.

Ask: What do you already know about this idea?

Expect: Area, squares, percent of a whole, and negative-number order.

If stuck: Ask which segment runs from center to circle.

Slide 589. Today

Topic: Circles and first previews
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

If stuck: Read the learning goal together.

Slide 590. Circumference from diameter | 5 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 257 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: Is the given segment a radius or a diameter? Are we finding boundary length or area?

Expect: P1: 8π cm; about 25.12 cm

If stuck: P1: Mark the center. Split a diameter into two radii before using the area formula.

Slide 591. Circumference from radius | 3 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 257 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: P2: Is the given segment a radius or a diameter? Are we finding boundary length or area?

Expect: P2: Diameter 6 cm; circumference 6π cm; P3: 16π square cm; about 50.24 square cm

If stuck: P2: Mark the center. Split a diameter into two radii before using the area formula.

Slide 592. The model | 3 min

Topic: Circles and first previews
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: A diameter contains two radii.

If stuck: Ask which segment runs from center to circle.

Slide 593. Area from diameter | 2 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 258 · P4, P5

Say: Read P4 and P5. Give time to try before comparing methods.

Ask: P4: Is the given segment a radius or a diameter? Are we finding boundary length or area?

Expect: P4: 9π square cm; about 28.26 square cm; P5: Both ratios equal π

If stuck: P4: Mark the center. Split a diameter into two radii before using the area formula.

Slide 594. Percent increase and decrease | 5 min

Topic: Circles and first previews
Student: Day 065 · Workbook pp. 258, 259 · P6, P7

Say: Read P6 and P7. Give time to try before comparing methods.

Ask: P6: Which amount is the starting whole for this percent change?

Expect: P6: 25% increase; 20% decrease; P7: 75 tokens; the decrease uses the larger whole 100

If stuck: P6: Label the starting amount for each step. Compare the change with that starting amount.

Slide 595. Positive addend | 5 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 259 · P8

Say: Read P8. Give time to try before comparing methods.

Ask: P8: Where do you start, and which direction does the operation move?

Expect: P8: 3; −7; −4

If stuck: P8: Mark the starting point. Use one unit per step and keep track of direction.

Slide 596. Watch the thinking | 2 min

Topic: Circles and first previews
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 25π square m; about 78.5 square m

If stuck: Return to the matching unit in the model.

Slide 597. Your turn | 2 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 259 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 8; 8; 64

If stuck: Compare with the previous example without copying its numbers.

Slide 598. A claim to check | 10 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 260 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Radius = 7 cm, so area = 49π square cm.

If stuck: Mark the center and split the diameter into two equal radii.

Slide 599. Words and meaning | 2 min

Topic: Circles and first previews
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: A radius of 3 cm gives diameter 6 cm. Diameter = 2 × radius. C = πd π is about 3.14, not exactly 3.14.

If stuck: Read one definition and draw its example.

Slide 600. In your own words | 1 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 260 · S

Say: Use your own words and one example.

Ask: Partner compare: How can the units tell us which circle formula fits?

Expect: Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

If stuck: Point to the model and name what each part represents.

Slide 601. Exit ticket | 5 min

Topic: Circles and first previews
Student: Day 065 · Workbook p. 261 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: Radius 6 cm; area 36π square cm; about 113.04 square cm

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 602. The week ahead | 5 min

Topic: Circles and first previews
Student: Days 066-070 · Workbook pp. 268-277

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 603. Final mixed review | 15 min

Topic: Circles and first previews
Student: Day 065 · Workbook pp. 262-263 · Final mixed review

Say: Use the printed check and keep the key out of view.

Ask: Read each question before starting.

Expect: Score by skill using the separate answer key.

If stuck: Let the student skip and return. Record unattempted items separately.

Slide 604. Tutor reference

Topic: Circles and first previews

Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: Radius 6 cm; area 36π square cm; about 113.04 square cm

Got it: Uses radius 6 and gives 36π square cm and about 113.04 square cm.

Almost there: Halves the diameter and squares the radius but makes one arithmetic or unit slip.

Needs re-teaching: Uses 12 as the radius, uses circumference, or treats 3.14 as an exact value.

PowerPoint answer review

After Friday, outside the weekend lesson time. Say: Check one completed item at a time. Ask: Where does your method first differ from this one? Expect: The answers and worked reasoning printed in the complete keys below. If stuck: Rebuild the quantity with its model, then have the student repair the first incorrect step.

Keep these slides out of view during independent work. The index below follows the answer-review slides in order; every answer is also printed in the complete keys.

Slide 605. Week 10 answers

This is the final answer section. Revisit any skill that needs more practice.

Slide 606. Scale and proportional relationships (1/4)

Items: S19:R1; S19:R2; S19:R3; S19:R4. Follow along: Day 064 · Workbook p. 251 · Items R1, R2, R3, R4

Slide 607. Scale and proportional relationships (2/4)

Items: S19:S; S19:P1; S19:P2; S19:P3. Follow along: Day 064 · Workbook pp. 251, 252 · Items S, P1, P2, P3

Slide 608. Scale and proportional relationships (3/4)

Items: S19:P4; S19:P5; S19:P6; S19:P7; S19:P8. Follow along: Day 064 · Workbook pp. 253, 254 · Items P4, P5, P6, P7, P8

Slide 609. Scale and proportional relationships (4/4)

Items: S19:F; S19:D; S19:E. Follow along: Day 064 · Workbook pp. 254, 255, 256 · Items F, D, E

Slide 610. Circles and first previews (1/3)

Items: S20:P1; S20:P2; S20:P3; S20:P4; S20:P5. Follow along: Day 065 · Workbook pp. 257, 258 · Items P1, P2, P3, P4, P5

Slide 611. Circles and first previews (2/3)

Items: S20:P6; S20:P7; S20:P8; S20:F; S20:D. Follow along: Day 065 · Workbook pp. 258, 259, 260 · Items P6, P7, P8, F, D

Slide 612. Circles and first previews (3/3)

Items: S20:S; S20:E. Follow along: Day 065 · Workbook pp. 260, 261 · Items S, E

Slide 613. Final mixed review (1/2)

Items: W10_Final_Mixed_Review:1; W10_Final_Mixed_Review:2; W10_Final_Mixed_Review:3; W10_Final_Mixed_Review:4; W10_Final_Mixed_Review:5. Follow along: Day 065 · Workbook p. 262 · Items 1, 2, 3, 4, 5

Slide 614. Final mixed review (2/2)

Items: W10_Final_Mixed_Review:6; W10_Final_Mixed_Review:7; W10_Final_Mixed_Review:8; W10_Final_Mixed_Review:9; W10_Final_Mixed_Review:10. Follow along: Day 065 · Workbook p. 263 · Items 6, 7, 8, 9, 10

Slide 615. Study guide: Try it

Items: W10_Study_Try:1; W10_Study_Try:2; W10_Study_Try:3; W10_Study_Try:4; W10_Study_Try:5. Follow along: Day 065 · Workbook p. 267 · Items 1, 2, 3, 4, 5

Slide 616. Daily practice (1/3)

Items: Day066_Practice:1; Day066_Practice:2; Day066_Practice:3; Day066_Practice:4; Day066_Practice:5. Follow along: Day 066 · Workbook p. 268 · Items 1, 2, 3, 4, 5

Slide 617. Daily practice (2/3)

Items: Day066_Practice:6; Day066_Practice:7; Day066_Practice:8; Day066_Practice:9; Day066_Practice:10. Follow along: Day 066 · Workbook pp. 268, 269 · Items 6, 7, 8, 9, 10

Slide 618. Daily practice (3/3)

Items: Day066_Practice:11; Day066_Practice:12; Day066_Practice:13; Day066_Practice:14; Day066_Practice:15. Follow along: Day 066 · Workbook p. 269 · Items 11, 12, 13, 14, 15

Slide 619. Daily practice (1/3)

Items: Day067_Practice:1; Day067_Practice:2; Day067_Practice:3; Day067_Practice:4; Day067_Practice:5. Follow along: Day 067 · Workbook p. 270 · Items 1, 2, 3, 4, 5

Slide 620. Daily practice (2/3)

Items: Day067_Practice:6; Day067_Practice:7; Day067_Practice:8; Day067_Practice:9; Day067_Practice:10. Follow along: Day 067 · Workbook pp. 270, 271 · Items 6, 7, 8, 9, 10

Slide 621. Daily practice (3/3)

Items: Day067_Practice:11; Day067_Practice:12; Day067_Practice:13; Day067_Practice:14; Day067_Practice:15. Follow along: Day 067 · Workbook p. 271 · Items 11, 12, 13, 14, 15

Slide 622. Daily practice (1/3)

Items: Day068_Practice:1; Day068_Practice:2; Day068_Practice:3; Day068_Practice:4; Day068_Practice:5. Follow along: Day 068 · Workbook p. 272 · Items 1, 2, 3, 4, 5

Slide 623. Daily practice (2/3)

Items: Day068_Practice:6; Day068_Practice:7; Day068_Practice:8; Day068_Practice:9; Day068_Practice:10. Follow along: Day 068 · Workbook pp. 272, 273 · Items 6, 7, 8, 9, 10

Slide 624. Daily practice (3/3)

Items: Day068_Practice:11; Day068_Practice:12; Day068_Practice:13; Day068_Practice:14; Day068_Practice:15. Follow along: Day 068 · Workbook p. 273 · Items 11, 12, 13, 14, 15

Slide 625. Daily practice (1/4)

Items: Day069_Practice:1; Day069_Practice:2; Day069_Practice:3; Day069_Practice:4; Day069_Practice:5. Follow along: Day 069 · Workbook p. 274 · Items 1, 2, 3, 4, 5

Slide 626. Daily practice (2/4)

Items: Day069_Practice:6; Day069_Practice:7; Day069_Practice:8; Day069_Practice:9; Day069_Practice:10. Follow along: Day 069 · Workbook pp. 274, 275 · Items 6, 7, 8, 9, 10

Slide 627. Daily practice (3/4)

Items: Day069_Practice:11; Day069_Practice:12; Day069_Practice:13; Day069_Practice:14; Day069_Practice:15. Follow along: Day 069 · Workbook p. 275 · Items 11, 12, 13, 14, 15

Slide 628. Daily practice (4/4)

Items: Day069_Practice:16; Day069_Practice:17; Day069_Practice:18. Follow along: Day 069 · Workbook p. 275 · Items 16, 17, 18

Slide 629. Weekly check (1/2)

Items: Day070_Practice:1; Day070_Practice:2; Day070_Practice:3; Day070_Practice:4; Day070_Practice:5. Follow along: Day 070 · Workbook p. 276 · Items 1, 2, 3, 4, 5

Slide 630. Weekly check (2/2)

Items: Day070_Practice:6; Day070_Practice:7; Day070_Practice:8; Day070_Practice:9; Day070_Practice:10. Follow along: Day 070 · Workbook p. 277 · Items 6, 7, 8, 9, 10

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 066. Monday | pages 268-269

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Scale, proportion and circles: items 11, 12, 13, 14. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 067. Tuesday | pages 270-271

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Scale, proportion and circles: items 11, 12, 13, 14. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 068. Wednesday | pages 272-273

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 15. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 069. Thursday | pages 274-275

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Data and statistics: items 11, 12, 13, 14. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 070. Friday | pages 276-277

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Scale and proportional relationships

Day 064, student workbook pages 251-256

R1 [G-AREA] | student p. 251

Answer: 16 square cm

Reasoning: Multiply the two perpendicular side lengths.

R2 [N-LCM] | student p. 251

Answer: LCM 12

Reasoning: List positive multiples until the lists first meet.

R3 [S-DISPLAY] | student p. 251

Answer: 2

Reasoning: Count every occurrence of 4.

R4 [G-SA] | student p. 251

Answer: 60 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 48.

Watch for 54: Omits one triangular end.

S [EXPLAIN] | student p. 251

Answer: Answers vary. A scaled copy multiplies every length by the same factor. Proportional relationships have the form $y = kx$.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [P-SCALE] | student p. 252

Answer: 8 cm by 12 cm; area 24 to 96 square cm, factor 4

Reasoning: Both dimensions double, so four originals fit in the copy.

Watch for Area 48 square cm: Applies the length factor once to area.

P2 [P-SCALE] | student p. 252

Answer: 3, 4, 5 cm; corresponding angles stay equal

Reasoning: Multiply every length by one half. Matching sides have the same ratio.

Watch for 5, 7, 9 cm: Subtracts one from each length instead of multiplying.

P3 [P-MAP] | student p. 252

Answer: 90 km

Reasoning: Each map centimeter represents 25 km, so multiply 3.6 × 25.

Watch for 0.144 km: Reverses map and actual units.

P4 [P-SCALE] | student p. 253

Answer: 45 square cm

Reasoning: There are three times as many units in each of two directions: factor 9.

Watch for 15 square cm: Uses the length factor for area.

P5 [P-PROP] | student p. 253

Answer: k = 2.5; y = 2.5x

Reasoning: Each y divided by its x gives 2.5.

Watch for k = 0.4: Finds x per y instead of y per x.

P6 [P-PROP] | student p. 253

Answer: A: y = 2x. B is not proportional because it starts at 3 when x = 0.

Reasoning: Both have equal increases, but only A has zero output for zero input.

Watch for Both, because both increase by 2: Mistakes constant differences for a constant ratio.

P7 [P-PROP] | student p. 254

Answer: c = 3m + 4; not proportional

Reasoning: At zero miles the cost is $4. Cost per mile changes as the fee is spread out.

Watch for c = 7m: Treats the starting fee as a per-mile charge.

P8 [P-PROP] | student p. 254

Answer: L = 1.5t, or L = $(\frac{3}{2})$t

Reasoning: Double the half-minute amount to get 1.5 liters per minute.

Watch for L = $(\frac{2}{3})$t: Reverses liters per minute and minutes per liter.

F [P-PROP] | student p. 254

Answer: 4; 3.5; 3.5

Reasoning: Use the worked example in the slides to check each step.

D [P-SCALE] | student p. 255

Answer: The area is multiplied by 9 because both dimensions are tripled.

Reasoning: Applies one length multiplier to a two-dimensional measure. Tile a 3 by 3 array of original rectangles.

E [P-PROP] | student p. 256

Answer: No. Ratios 5, $\frac{11}{3}$, $\frac{17}{5}$ differ. The rule y = 3x + 2 gives y = 2 at zero.

Reasoning: The change in y is always 6 when x increases by 2, but the ratio y to x is not constant.

Watch for Yes, y increases by 6 each time: Uses additive pattern instead of a constant multiplier.

Circles and first previews

Day 065, student workbook pages 257-261

P1 [G-CIRCLE] | student p. 257

Answer: 8π cm; about 25.12 cm

Reasoning: $C = \pi d = 8\pi$. Approximate with 8×3.14.

Watch for 64π cm: Uses a squared length for a boundary.

P2 [G-CIRCLE] | student p. 257

Answer: Diameter 6 cm; circumference 6π cm

Reasoning: Double the radius before multiplying by π.

Watch for 3π cm: Uses the radius as the diameter.

P3 [G-CIRCLE] | student p. 257

Answer: 16π square cm; about 50.24 square cm

Reasoning: $A = \pi r^2$. Square the radius first, then multiply by π.

Watch for 8π square cm: Uses circumference as area.

P4 [G-CIRCLE] | student p. 258

Answer: 9π square cm; about 28.26 square cm

Reasoning: The radius is 3 cm, so $A = \pi \times 3^2 = 9\pi$.

Watch for 36π square cm: Squares the diameter.

P5 [G-CIRCLE] | student p. 258

Answer: Both ratios equal π

Reasoning: Each circumference is π times its diameter. Dividing by the diameter leaves π.

Watch for The larger circle has the larger ratio: Confuses a larger circumference with a larger ratio.

P6 [P-CHANGE] | student p. 258

Answer: 25% increase; 20% decrease

Reasoning: The same change of 20 is compared with different starting amounts.

Watch for 25% for both: Uses the first starting value for both changes.

P7 [P-CHANGE] | student p. 259

Answer: 75 tokens; the decrease uses the larger whole 100

Reasoning: First add 20. Then subtract 25 from 100. The percent applies to a new starting amount.

Watch for 80 tokens: Assumes equal percentages are equal amounts.

P8 [N-ADD] | student p. 259

Answer: 3; -7; -4

Reasoning: A positive addend moves right. A negative addend or subtracting a positive moves left.

Watch for 7 for $-2 + 5$: Adds distances without direction.

F [G-CIRCLE] | student p. 259

Answer: 8; 8; 64

Reasoning: Use the worked example in the slides to check each step.

D [G-CIRCLE] | student p. 260

Answer: Radius = 7 cm, so area = 49π square cm.

Reasoning: Uses diameter where the formula requires radius. Mark the center and split the diameter into two equal radii.

S [EXPLAIN] | student p. 260

Answer: Answers vary. Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

Reasoning: Accept an accurate explanation supported by a matching example.

E [G-CIRCLE] | student p. 261

Answer: Radius 6 cm; area 36π square cm; about 113.04 square cm

Reasoning: Half of 12 is 6. $A = \pi \times 6^2 = 36\pi$.

Watch for 144π square cm: Squares the diameter instead of the radius.

Final mixed review

Day 065, student workbook pages 262-263

Score by skill, including the requested reasoning. Use 80% or more as secure, 50-79% as developing, and below 50% as a review priority. Revisit a missed model, then check again with a different problem. This tutoring check does not determine school placement.

1 [F-COMP] | student p. 262

Answer: $\frac{5}{8} > \frac{3}{5}$

Reasoning: Rename as twenty-five fortieths and twenty-four fortieths.

Watch for $\frac{5}{8} < \frac{3}{5}$: Compare equal-sized parts, not the original denominator counts.

2 [R-TABLE] | student p. 262

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

3 [R-RATE] | student p. 262

Answer: 4 labels per minute; $\frac{1}{4}$ minute per label

Reasoning: Divide labels by minutes for the first rate. Reverse for the second.

4 [F-DIV] | student p. 262

Answer: $\frac{5}{24}$ liter

Reasoning: Split each sixth into four equal parts. Each cup gets five twenty-fourths.

5 [D-DIV] | student p. 262

Answer: 0.225

Reasoning: Use long division into tenths, hundredths, and thousandths.

6 [E-DIST] | student p. 263

Answer: 2x + 14

Reasoning: Multiply x and 7 by 2.

Watch for 2x + 7: Multiplies only the first term.

7 [G-SA] | student p. 263

Answer: 60 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 48.

Watch for 54: Omits one triangular end.

8 [N-OPP] | student p. 263

Answer: 6

Reasoning: Reflect the point across zero.

9 [S-IQR] | student p. 263

Answer: Q1 2.5; Q3 9.5; IQR 7

Reasoning: Split the ordered list into equal-count halves. Subtract the two half-medians.

10 [G-CIRCLE] | student p. 263

Answer: 3π cm; about 9.42 cm

Reasoning: Use C = πd.

Study guide: Try it

Day 065, student workbook pages 264-267

1 [P-SCALE] | student p. 267

Answer: 6 cm and 8 cm

Reasoning: Multiply both lengths by the same factor.

2 [P-SCALE] | student p. 267

Answer: 15 cm and 18 cm

Reasoning: Multiply both lengths by the same factor.

3 [G-CIRCLE] | student p. 267

Answer: 3π cm; about 9.42 cm

Reasoning: Use C = πd.

4 [G-CIRCLE] | student p. 267

Answer: 5π cm; about 15.70 cm

Reasoning: Use C = πd.

5 [G-CIRCLE] | student p. 267

Answer: 7π cm; about 21.98 cm

Reasoning: Use C = πd.

Daily practice

Day 066, student workbook pages 268-269

1 [N-FACT] | student p. 268

Answer: 36

2 [N-FACT] | student p. 268

Answer: 84

3 [N-FACT] | student p. 268

Answer: 48

4 [N-FACT] | student p. 268

Answer: 108

5 [F-SCALE] | student p. 268

Answer: Larger; 8

Reasoning: Four thirds takes one whole six plus one third of six.

6 [R-RATE] | student p. 268

Answer: 0.65 dollars per marker

Reasoning: Use the named units, calculate, then check with the inverse operation.

7 [D-SUB] | student p. 268

Answer: 8.714

Reasoning: Write 10.000 and regroup through the zero columns.

8 [G-SA] | student p. 268

Answer: 84 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 72.

Watch for 78: Omits one triangular end.

9 [N-ORDER] | student p. 269

Answer: −9, −4, 0

Reasoning: Read the numbers from left to right on a number line.

10 [S-MAD] | student p. 269

Answer: 2

Reasoning: Mean 4. Average the distances from 4.

Watch for 0: Averages signed differences instead of distances.

11 [P-SCALE] | student p. 269 | Built-in review: Scale, proportion and circles

Answer: 4 cm and 6 cm

Reasoning: Multiply both lengths by the same factor.

12 [P-MAP] | student p. 269 | Built-in review: Scale, proportion and circles

Answer: 80 km

Reasoning: 4 map centimeters represent 4 groups of 20 km.

13 [P-SCALE] | student p. 269 | Built-in review: Scale, proportion and circles

Answer: 5 square cm

Reasoning: Multiply area by the square of the scale factor.

14 [P-PROP] | student p. 269 | Built-in review: Scale, proportion and circles

Answer: k = 1.5

Reasoning: Divide the output by the input.

15 [P-SCALE] | student p. 269

Answer: The area is multiplied by 9 because both dimensions are tripled.

Watch for Area triples: Applies one length multiplier to a two-dimensional measure.

Daily practice

Day 067, student workbook pages 270-271

1 [F-EQUIV] | student p. 270

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 270

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 270

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 270

Answer: 0.125; 12.5%

5 [F-LINE] | student p. 270

Answer: At $1\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

6 [R-COMP] | student p. 270

Answer: Mix A

Reasoning: Use equivalent ratios with the same water amount, then compare juice.

Watch for Mix B: Comparing only the juice counts ignores the amount of water.

7 [F-DIV] | student p. 270

Answer: $4\ \frac{2}{3}$

Reasoning: Rename the mixed number as an improper fraction, then count groups of three fourths.

Watch for $2\ \frac{5}{8}$: Multiplying by the divisor answers a different question.

8 [E-MULT] | student p. 270

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

9 [N-ABS] | student p. 271

Answer: 0 units from zero

Reasoning: Count the unit steps between the number and zero.

10 [S-MEDIAN] | student p. 271

Answer: 5

Reasoning: Order: 2, 4, 6, 8. Use the middle value or mean of two middles.

11 [G-CIRCLE] | student p. 271 | Built-in review: Scale, proportion and circles

Answer: 3π cm; about 9.42 cm

Reasoning: Use C = πd.

12 [G-CIRCLE] | student p. 271 | Built-in review: Scale, proportion and circles

Answer: 4π cm; about 12.56 cm

Reasoning: Double the radius for the diameter, then multiply by π.

13 [G-CIRCLE] | student p. 271 | Built-in review: Scale, proportion and circles

Answer: 4π square cm; about 12.56 square cm

Reasoning: Square the radius, then multiply by π.

14 [G-CIRCLE] | student p. 271 | Built-in review: Scale, proportion and circles

Answer: 4π square cm; about 12.56 square cm

Reasoning: Halve the diameter, square the radius, then multiply by π.

Watch for 16π square cm: Squares the diameter.

15 [G-CIRCLE] | student p. 271

Answer: Radius = 7 cm, so area = 49π square cm.

Watch for 196π square cm: Uses diameter where the formula requires radius.

Daily practice

Day 068, student workbook pages 272-273

1 [N-SCALE] | student p. 272

Answer: 122

2 [N-SCALE] | student p. 272

Answer: 12

3 [N-SCALE] | student p. 272

Answer: 25

4 [N-SCALE] | student p. 272

Answer: 36

5 [F-COMP] | student p. 272

Answer: $\frac{2}{4} > \frac{2}{5}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{2}{4} < \frac{2}{5}$: A larger denominator makes smaller equal parts, not a larger fraction.

6 [R-WHOLE] | student p. 272

Answer: 10 yellow; 25 purple

Reasoning: Add ratio parts: 2 + 5 = 7. Divide the total by 7, then multiply each ratio part.

7 [P-PART] | student p. 272

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

8 [F-DIV] | student p. 272

Answer: $\frac{2}{5}$

Reasoning: Two and one half is five halves. Five halves times two fifths equals one.

9 [D-ADD] | student p. 273

Answer: 6.528

Reasoning: Line up decimal points. Add digits in the same place-value columns.

10 [E-MULT] | student p. 273

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

11 [G-SA] | student p. 273

Answer: 58 square cm

Reasoning: Two faces of each type: 2 × (4 + 5 + 20).

Watch for 20: Uses volume.

12 [E-INEQ] | student p. 273

Answer: x < 5; no

Reasoning: Use a closed circle if equality is allowed; otherwise use an open circle.

13 [S-SPREAD] | student p. 273

Answer: Both means 4. Range A 2; range B 6. B is more spread out.

Reasoning: Find the center, then compare the endpoint gaps.

14 [G-CIRCLE] | student p. 273

Answer: 3π cm; about 9.42 cm

Reasoning: Use $C = \pi d$.

15 [P-SCALE] | student p. 273 | Built-in review: Weekly puzzle

Answer: New drawing: 1.5 cm by 2 cm. Actual area: 1,200 square m.

Reasoning: The garden is 30 m by 40 m. Divide those lengths by 20 for the new drawing. The garden itself stays the same.

Watch for 600 square m: Changes the actual area when only the drawing scale changes.

Daily practice

Day 069, student workbook pages 274-275

1 [N-EST] | student p. 274

Answer: 1200

2 [N-EST] | student p. 274

Answer: 13

3 [N-EST] | student p. 274

Answer: About 100

4 [N-EST] | student p. 274

Answer: About 20

5 [R-EQUIV] | student p. 274

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

6 [P-PART] | student p. 274

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

7 [F-DIV] | student p. 274

Answer: $2\ \frac{1}{2}$; stories vary

Reasoning: Example: how many one-third-meter lengths fit in five sixths meter, allowing part of a length?

8 [D-DIV] | student p. 274

Answer: 1.2

Reasoning: Ninety-six tenths shared among eight gives twelve tenths.

9 [E-ADD] | student p. 275

Answer: x = 17

Reasoning: Subtract 14 from both sides.

10 [G-AREA] | student p. 275

Answer: 21 square cm

Reasoning: Multiply the two perpendicular side lengths.

11 [S-MEAN] | student p. 275 | Built-in review: Data and statistics

Answer: 4

Reasoning: Total 16 divided by 4 values gives 4.

12 [S-MEDIAN] | student p. 275 | Built-in review: Data and statistics

Answer: 6

Reasoning: Order: 2, 3, 5, 7, 9, 10. Use the middle value or mean of two middles.

13 [S-RANGE] | student p. 275 | Built-in review: Data and statistics

Answer: 10

Reasoning: 10 − 0 = 10.

14 [S-MAD] | student p. 275 | Built-in review: Data and statistics

Answer: $\frac{3}{2}$

Reasoning: Mean 5. Average the distances from 5.

Watch for 0: Averages signed differences instead of distances.

15 [P-MAP] | student p. 275

Answer: 100 km

Reasoning: 2 map centimeters represent 2 groups of 50 km.

16 [G-CIRCLE] | student p. 275

Answer: Radius = 7 cm, so area = 49π square cm.

Watch for 196π square cm: Uses diameter where the formula requires radius.

17 [N-SIGNED] | student p. 275

Answer: −12; 4

18 [R-PERCENT] | student p. 275

Answer: 25%

Weekly check

Day 070, student workbook pages 276-277

1 [F-MULT] | student p. 276

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

2 [R-GRAPH] | student p. 276

Answer: Yes

Reasoning: Every vertical value is twice the horizontal value.

3 [P-PART] | student p. 276

Answer: 8 stickers

Reasoning: Twenty-five percent is one fourth. Divide the whole by 4.

4 [F-DIV] | student p. 276

Answer: 1 meters

Reasoning: Divide area by width. Check that length times width equals area.

Watch for $\frac{9}{16}$: Multiplying by the divisor answers a different question.

5 [D-MULT] | student p. 276

Answer: 0.0100

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.1000: The decimal point is one place too far right. Check the product against both factors.

6 [E-DIST] | student p. 277

Answer: 3x + 18

Reasoning: Multiply x and 6 by 3.

Watch for 3x + 6: Multiplies only the first term.

7 [G-SA] | student p. 277

Answer: 58 square cm

Reasoning: Two faces of each type: 2 × (4 + 5 + 20).

Watch for 20: Uses volume.

8 [N-OPP] | student p. 277

Answer: 1

Reasoning: Reflect the point across zero.

9 [S-MAD] | student p. 277

Answer: 3

Reasoning: Mean 6. Average the distances from 6.

Watch for 0: Averages signed differences instead of distances.

10 [P-SCALE] | student p. 277

Answer: The area is multiplied by 9 because both dimensions are tripled.

Watch for Area triples: Applies one length multiplier to a two-dimensional measure.